

IVACS V-TRACE: Vehicle Trust, Route & Evidence Engine

Problem Statement: OD-08 — License Plate Detection, Recognition & Vehicle Identity Engine

Report Updated: 2026-09-18 11:22:28	Phase: Vehicle Visual Fingerprint Verified	Policy: Zero LLM / CV-Native
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1. System Component Status Audit

Component / Subsystem	Verification Status
Backend Server (FastAPI REST App & Background Demo Worker)	PASS
Video Ingestion (MP4 Sequential Stream & Configurable Sampling)	PASS
Vehicle Detection (Ultralytics YOLOv8 Nano Pretrained)	PASS
Plate Localization (Bumper ROI & Vehicle Association)	PASS
OCR Character Recognition (EasyOCR Dual Text Normalization)	PASS
Vehicle Visual Fingerprint (ResNet18 512-dim Normalized Embedding)	PASS
Identity Matching & Decision Rules (Rules A through E)	PASS
Database Persistence (SQLite: detections, vehicle_identities, observations)	PASS
Evidence Image Storage (Frame, Vehicle, Plate, Annotated Crops)	PASS
REST API Layer (Health, Cameras, Demo, Detections, Vehicles, Identity-Events)	PASS
Automated Test Suite (14/14 Tests Passed - 100% Pass Rate)	PASS

2. Team Member Role Allocation (CodeAvengers Team)

Member	GitHub / Email	Assigned Subsystem Role
Jone Kavish (Lead)	jonekavish-dot jonekavish@gmail.com	Backend Architecture, Overall Team Management, EasyOCR Pipeline, REST Endpoints, Frame Orchestrator
K.V. Pranesh (Member-1)	kvpranesh kvpranesh49@gmail.com	Computer Vision & Vehicle Detection Engine (YOLOv8n Inference, Multi-Class COCO Vehicle Filtering)
Gowshik Gunal (Member-2)	gowshikgunal22 gowshikgunal@gmail.com	License Plate Localization Engine (Bumper ROI Analysis, Vehicle Association, Image Preprocessing)
Dinesh Balu (Member-3)	dineshbalu7f-glitch dineshbalu7.f@gmail.com	Video Source Abstraction (MP4/RTSP/Webcam Streams), Frame Sampling, Evidence Storage & SQLite Persistence

3. Identity Engine Decision Rules Audit

Rule	Condition / Observation	System Analytical Event Signal
Rule A	Same Plate + High Visual Similarity (≥ 0.85)	SAME_VEHICLE (Confirmed consistent identity)
Rule B	Same Plate + Low Visual Similarity (< 0.85)	POSSIBLE_IDENTITY_MISMATCH (Appearance conflict)
Rule C	Different Plate + High Visual Similarity (≥ 0.85)	POSSIBLE_PLATE_SWAP (Plate swap anomaly)
Rule D	Plate Unreadable + High Visual Similarity (≥ 0.85)	PLATE_UNREADABLE_VEHICLE_MATCH (Continuity)
Rule E	No Historical Match Found	NEW_VEHICLE (New identity registration)